

SAGUACHE, CO, USA

WMO: 722333

Lat: 38.097N

Lon: 106.169W

Elev: 2385

StdP: 75.76

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03069

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-16.2	-23.7	0.6	-9.5	-22.2	0.7	-8.9	14.7	0.8	12.6	-1.0	2.7	330	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	30.0	12.1	28.8	11.9	27.4	11.9	15.1	24.0	14.4	23.0	13.8	22.5	4.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.6	12.2	16.6	12.1	11.8	16.2	11.0	11.0	16.6	50.9	24.2	49.0	23.5	47.1	21.5	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.9	9.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-5.5	-1.8	3.7	7.0	11.0	16.8	19.5	18.1	14.7	8.2	2.2	-5.2		
	DBStd	9.27	4.28	4.34	3.88	3.49	3.71	2.94	2.10	2.11	2.60	3.61	4.16	5.41		
	HDD10.0	1923	480	329	196	99	34	1	0	0	2	75	234	473		
	HDD18.3	4068	738	563	453	340	229	62	12	31	112	314	484	731		
	CDD10.0	984	0	0	1	9	64	205	293	251	142	19	0	0		
	CDD18.3	89	0	0	0	0	0	16	47	23	3	0	0	0		
	CDH23.3	1829	0	0	0	2	55	523	714	407	128	2	0	0		
	CDH26.7	429	0	0	0	0	4	135	197	83	10	0	0	0		
(14) Wind	WSAvg	3.7	3.1	3.9	4.4	4.5	4.1	4.0	3.4	3.1	2.9	3.8	3.7	3.3		
(15) Precipitation	PrecAvg	214	6	6	11	14	17	15	37	41	26	18	11	8		
	PrecMax	375	26	32	32	69	50	60	92	86	80	71	42	50		
	PrecMin	133	0	0	0	0	0	0	4	10	3	0	0	0		
	PrecStd	53	6	6	8	14	12	12	23	19	18	18	10	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	12.5	19.9	22.8	27.2	32.0	32.6	30.2	27.8	22.9	17.2	10.3		
		MCWB	0.9	2.9	5.8	7.0	9.4	11.7	12.6	12.3	11.0	9.1	5.4	2.4		
	2%	DB	6.1	10.8	17.4	20.0	25.1	29.9	30.2	28.8	26.2	21.1	14.9	7.4		
		MCWB	-0.4	2.1	4.6	6.4	8.7	11.2	12.8	12.3	10.4	8.3	4.7	0.6		
	5%	DB	3.7	7.7	15.1	17.8	22.7	28.0	28.9	27.4	25.0	18.7	12.5	5.2		
		MCWB	-1.1	1.0	3.7	5.3	7.9	10.6	12.8	12.6	10.7	7.3	3.4	-0.5		
	10%	DB	1.4	5.9	12.5	16.2	21.1	27.0	27.4	26.0	22.9	17.1	10.2	2.7		
		MCWB	-1.9	-0.2	2.8	4.7	7.6	10.4	12.8	12.6	9.9	6.7	2.6	-1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	4.1	6.2	8.0	10.8	14.3	16.2	16.0	14.0	11.2	6.3	3.1		
		MCDB	6.4	10.9	16.6	18.9	21.1	23.6	25.6	23.9	23.4	18.8	14.9	7.4		
	2%	WB	0.7	2.7	5.1	7.2	9.8	13.2	15.2	15.1	12.9	9.6	5.1	1.5		
		MCDB	3.4	8.6	15.9	17.6	20.8	24.4	24.7	23.1	21.4	17.5	13.7	5.9		
	5%	WB	-0.4	1.5	4.1	6.2	9.0	12.4	14.6	14.4	12.1	8.4	4.0	0.1		
		MCDB	2.9	7.0	14.5	16.8	19.8	23.9	24.0	21.9	20.5	16.4	12.0	4.4		
	10%	WB	-1.6	0.3	3.0	5.3	8.2	11.4	14.0	13.8	11.3	7.2	2.8	-1.1		
		MCDB	1.6	5.0	11.4	14.8	18.8	23.0	23.3	21.5	19.9	15.6	9.5	2.8		
(35) Mean Daily Temperature Range	MDBR		12.4	12.9	14.9	15.6	16.1	18.2	16.7	15.8	17.0	15.7	14.3	12.6		
	5% DB	MCDBR	13.7	14.7	18.4	19.3	19.3	20.2	18.8	17.7	18.7	18.5	16.9	14.4		
		MCWBR	9.1	8.4	9.0	9.0	8.4	8.5	6.6	6.5	8.0	9.1	9.2	9.0		
	5% WB	MCDBR	11.6	13.5	17.1	17.7	17.4	17.5	15.8	14.6	15.9	16.4	16.0	13.3		
		MCWBR	8.2	8.2	8.6	8.5	7.8	7.5	5.9	5.8	7.0	8.3	8.9	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.223	0.229	0.257	0.280	0.299	0.304	0.348	0.330	0.288	0.260	0.235	0.226		
	taud		2.574	2.573	2.537	2.507	2.510	2.502	2.434	2.459	2.589	2.637	2.625	2.567		
	Ebn at Noon		998	1028	1017	1001	979	970	924	937	968	977	975	970		
	Edh at Noon		70	81	94	103	105	106	113	108	89	76	67	66		
(44) All-Sky Solar Radiation	RadAvg		2.91	3.88	5.17	6.14	6.75	7.44	6.46	5.90	5.38	4.21	3.13	2.52		
	RadStd		0.15	0.23	0.37	0.35	0.51	0.43	0.35	0.27	0.31	0.28	0.18	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	+24

Nomenclature: See separate page